Project 3: Final Report

Identification of Risk Factors and 10-year Stroke Risk in the Framingham Heart Study

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1. Introduction

The Framingham Heart Study (**FHS**) is a long-term prospective study investigating cardiovascular disease among individuals in Framingham, Massachusetts. This dataset includes up to 24 years of biennial follow-up for 4,434 participants, with repeated measures of clinical, laboratory, and questionnaire data, as well as adjudicated cardiovascular events.

The primary objective of this analysis is to identify baseline risk factors associated with incident stroke within 10 years and to estimate 10-year stroke probability across clinically relevant risk profiles. Analyses were stratified by sex to account for known differences in cardiovascular risk.

A secondary aim was to describe changes in key risk factors across three examination periods to assess whether a time-varying covariate approach may be warranted in future analyses.

The purpose of this report is to answer the primary hypothesis that baseline risk factors are associated with time to stroke within 10 years of follow-up (H_0: HR = 1 vs. H_a: HR ≠ 1), and how does the 10-year probability of stroke vary by risk profile and sex among participants in the FHS.

2. Methods

2.1 Data Cleaning

The original FHS dataset contained 4,434 participants with up to 24 years of follow-up structured in long format with repeated biennial visits per participant. Analyses were restricted to baseline (Exam 1) and the first 10 years of follow-up. Participants with prior stroke were excluded, yielding a final sample of 4,402 individuals.

The primary outcome is time to first stroke, measured in days, within 10 years. Participants without stroke were censored at 10 years or last follow-up. A binary indicator of stroke within 10 years (yes/no), was also created which will be used for computing 10-year stroke probabilities by risk profile.

Systolic blood pressure (**SBP**) and total cholesterol were rescaled so that the reported model coefficients correspond to a 10-mmHg and a 10-mg/dL increments, respectively, to improve interpretability.

2.2 Data Analysis

To address the primary research question, sex-stratified Cox proportional hazards models were used to identify risk factors associated with time to stroke. The proportional hazards assumption was assessed using Schoenfeld residuals.

Candidate baseline risk factors included age, SBP, and diabetes, selected a priori based on discussions with primary investigator (**PI**), with total cholesterol, BMI, smoking status, BP medication use, and prevalent cardiovascular disease (**CHD**) considered using backward stepwise selection based on corrected Akaike's Information Criterion (AICc), Schwarz Bayesian Criterion (SBC) criteria. Clinical relevance informed final model selection.

To address the secondary research question, 10-year stroke probabilities were estimated from Cox models for five prespecified risk profiles at ages 40, 50, and 60. The scenarios included a baseline profile with no major risk factors, high BP alone, diabetes alone, diabetes with high BP, and diabetes with high BP and a current smoker. Predicted risk was calculated as one minus the estimated 10-year stroke-free survival probability from the sex-specific Cox models.

As an additional descriptive aim, risk factors were also summarized across three examination periods using means (SD) and proportions to assess changes over time.

All analyses used complete-case data under a missing at random assumption, where dropout depends on observed variables. This resulted in a final analytical sample of 2,378 women and 1,897 men. Statistical significance was assessed at a two-sided $\alpha = 0.05$ level. All analyses were conducted in R (version 4.5.3; Reassured Reassurer).

3. Results

Characteristics of the Framingham Heart Study Cohort

Characteristic	N	Overall N = 4,275[1,2]	Men N = 1,897[1,2]	Women N = 2,378[1,2]
Age	4,275	49 (42, 57)	49 (42, 57)	49 (43, 57)
Systolic BP	4,275	129 (117, 144)	129 (118, 141)	129 (116, 146)
Diabetes	4,275			
No		4,162 (97%)	1,841 (97%)	2,321 (98%)
Yes		113 (2.6%)	56 (3.0%)	57 (2.4%)
Total Cholesterol (mg/dL)	4,275	234 (206, 264)	231 (206, 259)	237 (206, 268)
Cardiovascular Disease	4,275			
No		4,098 (96%)	1,784 (94%)	2,314 (97%)
Yes		177 (4.1%)	113 (6.0%)	64 (2.7%)
BP Meds	4,275			
No		4,143 (97%)	1,857 (98%)	2,286 (96%)
Yes		132 (3.1%)	40 (2.1%)	92 (3.9%)
Smoker	4,275			
No		2,170 (51%)	750 (40%)	1,420 (60%)
Yes		2,105 (49%)	1,147 (60%)	958 (40%)
BMI	4,275	25.5 (23.1, 28.1)	26.1 (24.0, 28.3)	24.8 (22.5, 27.8)
Event (Stroke)	4,275			
No-Censored		4,170 (98%)	1,849 (97%)	2,321 (98%)
Yes		105 (2.5%)	48 (2.5%)	57 (2.4%)

[1] Median (Q1, Q3); n (%)

[2] Table includes only participants that have all baseline measurements (complete-cases)

Table 1. *Baseline characteristics of Framingham Heart Study participants, overall and stratified by sex during 10 years of follow-up.*

Baseline characteristics are shown in **Table 1.** Smoking and prevalent CHD were more common in men, while other risk factors were similar between sexes. Over 10 years, 111 stroke events

occurred among 2,472 women and 1,930 men, corresponding to a cumulative incidence of approximately 2.5%. Missingness was low (≈2.9%) and not expected to meaningfully impact results.

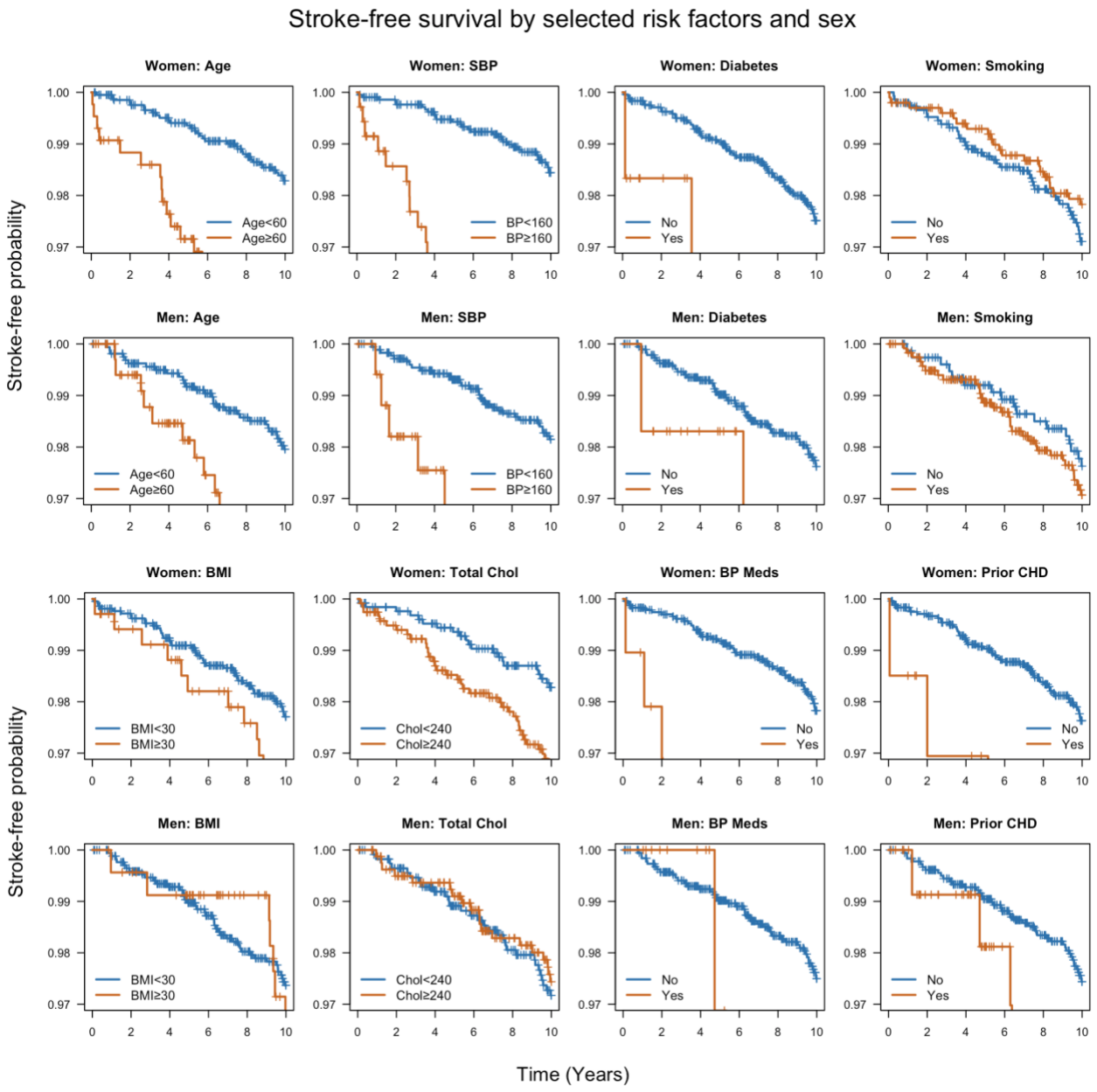

Figure 1. *Kaplan–Meier curves of stroke-free survival over 10 years, stratified by sex and baseline risk factors. Each panel displays survival probabilities for levels of the corresponding risk factor for women and men, with tick marks denoting censored observations.*

Kaplan–Meier curves, stratified by selected baseline risk factors, showed lower stroke-free survival among individuals with elevated SBP (≥160mmHg), diabetes, older age (≥60yrs), and prevalent CHD.

3.1 Risk Factors of Stroke

Stepwise procedures identified smoking as the only statistically significant predictor of stroke. However, prior literature, including the Framingham Stroke Risk Profile (*Wolf et al., 1991*), incorporated an interaction between SBP and BP medication use, as well as prevalent CHD, based on clinical relevance. These terms were therefore retained in the final sex-specific Cox models.

Hazard Ratios and Percent Change in Stroke Hazard by Sex

	Women (N=2378)			Men (N=1897)		
Risk factor	**Events**	**HR (95% CI; P-val)**	**%Δ in Hazard**	**Events**	**HR (95% CI; P-val)**	**%Δ in Hazard**
Age (per 1 year)		1.08 (1.03, 1.12; p=<0.01)	+7.7%		1.07 (1.03, 1.11; p=<0.01)	+6.7%
Systolic BP (per 10 mmHg)		1.28 (1.17, 1.41; p=<0.01)	+28.0%		1.41 (1.26, 1.59; p=<0.01)	+41.4%
BP meds (Yes vs No)	10/96	1.77 (0.03, 121.95; p=0.79)	+76.7%	4/40	+[1]	+[1]
Diabetes (Yes vs No)	4/60	1.82 (0.66, 5.07; p=0.25)	+82.2%	7/59	5.06 (2.19, 11.68; p=<0.01)	+406.0%
Current smoker (Yes vs No)	21/1001	1.45 (0.82, 2.54; p=0.20)	+44.6%	32/1168	1.95 (1.04, 3.64; p=0.036)	+94.8%
Cardiovascular Disease (Yes vs No)	7/67	1.62 (0.68, 3.87; p=0.28)	+61.7%	5/120	1.00 (0.35, 2.90; p=0.997)	+0.2%
Total Chol (per 10 mg/dL)		0.98 (0.93, 1.04; p=0.61)	-1.5%		1.02 (0.96, 1.08; p=0.534)	+2.0%
SBP × BP meds interaction		0.99 (0.79, 1.25; p=0.95)	-0.8%		0.64 (0.43, 0.97; p=0.033)[1]	-35.7%[1]

[1] † Because the BP × BP medication interaction was significant in men, the effect of BP meds depends on BP and is not summarized as a single hazard ratio.

Table 2. *Sex-specific Cox proportional hazards models for 10-year stroke risk. Hazard ratios (95% CI; p-value) and percent change in hazard are shown for each risk factor. Continuous variables are scaled as indicated, and categorical variables are compared to their reference groups.*

In both sexes, older age and higher SBP were associated with higher risk of stroke at any given moment during follow-up period (**Table 2**). Each 10 mmHg increase in SBP was associated with approximately a 28% higher risk of stroke in women and 41% in men. In men, diabetes and smoking were also identified as risk factors, with diabetes being associated with an

approximately fivefold higher risk of stroke, although the small number of events (N=7) suggests uncertainty in this estimate. The interaction indicates that, at higher levels of SBP, men taking BP medication have a lower risk of stroke over time compared to those not on medication, suggesting that medication modifies the effect of SBP (***Supplementary Figure 1***).

Other predictors, mainly in women, were not statistically significant, likely reflecting limited power due to the relatively small number of stroke events within each stratum. For example, there were only seven events among women and five among men with prevalent CHD. Although CHD is a well-established risk factor for stroke, it was not statistically significant in this analysis.

The proportional hazards assumption was assessed using Schoenfeld residuals and was not violated for the final sex-stratified cox models (***Supplementary Table 1***).

3.2 10-year Probability of Stroke by Risk Profile

The estimated 10-year probabilities of stroke were low for individuals without any major risk factors, reflecting the relatively low incidence of stroke in this cohort over a 10-year period. Predicted risk increased with age and the presence of risk factors in both men and women. Within women, elevated BP (≥160mmHg) was associated with higher predicted stroke risk than diabetes across all age groups. In contrast, among men, diabetes was associated with higher predicted risk than elevated BP at each age. The highest risks were observed among individuals with multiple risk factors, particularly for men, where having diabetes, elevated BP, and being a current smoker resulted in the highest probabilities (***Table 3***).

Predicted 10-Year Probability of Stroke by Risk Profile and Sex						
	Women			Men		
Risk profile	**Age 40**	**Age 50**	**Age 60**	**Age 40**	**Age 50**	**Age 60**
Baseline	0.4% (0.1, 0.7)	0.8% (0.4, 1.3)	1.7% (0.8, 2.7)	0.3% (0.0, 0.6)	0.6% (0.2, 1.1)	1.2% (0.4, 2.1)
High BP	1.1% (0.2, 1.9)	2.2% (1.2, 3.3)	4.6% (2.8, 6.4)	1.3% (0.2, 2.5)	2.6% (1.0, 4.1)	4.9% (2.2, 7.5)
Diabetes	0.7% (0.0, 1.6)	1.5% (0.0, 3.2)	3.2% (0.0, 6.6)	1.7% (0.0, 3.5)	3.2% (0.3, 6.1)	6.1% (0.6, 11.3)
Smoker	0.6% (0.2, 1.0)	1.2% (0.6, 1.8)	2.5% (0.9, 4.1)	0.7% (0.2, 1.1)	1.3% (0.6, 1.9)	2.4% (1.0, 3.8)
Diabetes + High BP	1.9% (0.0, 4.3)	4.0% (0.0, 8.1)	8.3% (0.0, 16.0)	6.6% (0.0, 13.3)	12.3% (1.7, 21.8)	22.3% (5.0, 36.4)
Smoker + Diabetes + High BP	2.8% (0.0, 6.1)	5.8% (0.0, 11.7)	11.7% (0.0, 23.0)	12.5% (0.0, 24.0)	22.5% (2.8, 38.3)	38.8% (6.6, 59.9)

Table 3. *Predicted 10-year stroke risk (%) by age, sex, and risk profile from sex-specific Cox models with 95% CI. Other risk factors were held constant at reference levels (absent of disease).*

3.3 Time-varying Covariates

Risk factor levels and prevalence changed over time in both men and women. Systolic BP and diabetes prevalence increased over time, while smoking prevalence declined in both sexes (***Table 4***).

Risk Factors Across Examination Periods by Sex								
	Women				Men			
Period	**Age**	**SBP**	**Diabetes**	**Smoker**	**Age**	**SBP**	**Diabetes**	**Smoker**
Exam 1	50.0 (8.6)	133.8 (24.5)	2.5%	40.4%	49.8 (8.7)	131.7 (19.4)	3.0%	60.4%
Exam 2	55.7 (8.6)	138.1 (24.3)	3.6%	37.8%	55.1 (8.5)	135.5 (19.9)	4.4%	52.0%
Exam 3	60.9 (8.4)	140.9 (24.1)	7.1%	31.0%	60.3 (8.2)	139.3 (21.2)	8.7%	38.9%

Table 4. *Risk factor summaries across three examination periods in the Framingham Heart Study, stratified by sex. Continuous variables are presented as mean (SD), and categorical variables are presented as percentages. Estimates reflect cross-sectional summaries at each examination period.*

4. Discussion

High SBP (hypertension) and older age were identified as the primary risk factor for stroke in both sexes, consistent with findings from Wolf et al. Among men, smoking, diabetes and the

interaction between SBP and BP medication use were also identified as important predictors of stroke.

Although several additional risk factors, were not statistically significant, their estimated effects were generally consistent with prior literature. The lack of statistical significance is likely attributable to the relatively low number of stroke events.

Predicted 10-year stroke risk increased with both age and accumulation of risk factors, with the highest probabilities observed among individuals with multiple concurrent risk factors. In women, high BP had a greater impact on predicted risk than diabetes, while in men, it was the opposite.

Risk factor distributions were also observed to change across examination periods, with increases in SBP and diabetes prevalence and declines in smoking prevalence over time. These findings suggest that a time-varying covariate analysis may provide more accurate estimates in future work.

4.1 Limitations

Several limitations should be considered. First, the low number of stroke events limited statistical power, particularly for less prevalent risk factors and within sex-stratified analyses. Second, the use of baseline covariates may not capture changes in risk over time. Third, while model selection was guided by multiple information criteria, stepwise variable selection may introduce bias and overfitting, especially with correlated predictors. Finally, complete-case analysis may introduce bias if missingness is not at random.

4.2 Conclusions

In this cohort from the Framingham Heart Study, age and SBP were identified as the primary determinants of 10-year stroke risk in both men and women. While several additional risk factors were not statistically significant, their estimated effects were large, suggesting that limited statistical power may have influenced inference.

Predicted stroke risk increased with the accumulation of risk factors, particularly among men. Summary measures of risk factors changed over time, which suggests that incorporating time-varying covariates may improve estimation of stroke risk.

References

ChatGPT was used to check R code errors for some of the packages used and to grammar check (OpenAI, 2025).

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Wolf, P. A., D'Agostino, R. B., Belanger, A. J., & Kannel, W. B. (1991). Probability of stroke: A risk profile from the Framingham Study. *Stroke, 22*(3), 312–318.

Boehme, A. K., Esenwa, C., & Elkind, M. S. (2017). Stroke Risk Factors, Genetics, and Prevention. *Circulation research*, *120*(3), 472–495. https://doi.org/10.1161/CIRCRESAHA.116.308398

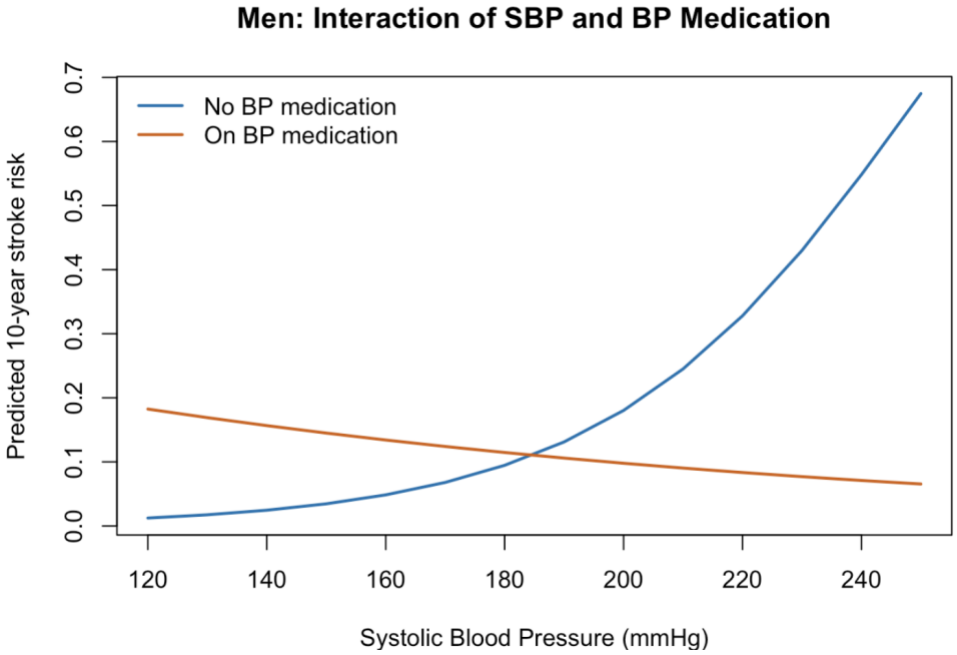
Reproducibility Statement

The statistical code used in this analysis is available on my GitHub under the Code directory at: https://github.com/rysummers/BIOS6624-Class/tree/main/Project3

Raw data are not tracked in this repository and is available upon request.

Link to data on local drive: /Users/ryan_summers/GitHub/BIOS6624-Class/Project3/DataRaw/frmgham2.csv

Supplemental



Supplementary Figure 1 *Predicted 10-year stroke risk by SBP and BP medication use in men. Risk increased sharply with SBP among men not on medication but was attenuated among treated men. The slight decrease in risk at higher SBP levels among men on medication likely reflects limited data and patients having a higher baseline SBP to begin with.*

Proportionality Test: Women				Proportionality Test: Men			
Variable	chisq	df	p	Variable	chisq	df	p
AGE	0.04	1	0.834	AGE	0.47	1	0.491
SYSBP_10	0.05	1	0.825	SYSBP_10	1.46	1	0.227
BPMEDS	3.12	1	0.078	BPMEDS	0.06	1	0.803
DIABETES	0.09	1	0.759	DIABETES	2.76	1	0.096
CURSMOKE	0.14	1	0.705	CURSMOKE	0.04	1	0.846
PREVCHD	0.00	1	0.951	PREVCHD	0.27	1	0.602
SYSBP_10:BPMEDS	2.62	1	0.106	SYSBP_10:BPMEDS	0.11	1	0.742
GLOBAL	4.14	7	0.764	GLOBAL	4.90	7	0.672

Supplementary Table 1 *Tests of the proportional hazards assumption for Cox models stratified by sex. Schoenfeld residual-based tests showed no evidence of violation of the proportional hazards assumption for any covariate or globally in either sex (all $p > 0.05$).*